RouteWise AI: Technical & Optimization Report

Multi-Objective Global Shipment Optimization Under Cost, Time & Risk Constraints

1. EXECUTIVE SUMMARY & OPERATIONAL CONTEXT

Global supply chains operate in an increasingly volatile environment characterized by escalating freight costs, port congestion, geopolitical choke-points, and stringent service level agreements (SLAs). Selecting optimal shipping routes requires balancing fundamentally competing objectives: minimizing direct monetary cost, accelerating transit days, and dampening exposure to delay and geopolitical instability.

RouteWise AI delivers an enterprise-grade decision-support optimization engine engineered in Python. The system evaluates multimodal route alternatives, enforces inviolable hard business constraints, computes pareto-efficient weighted composite scores via min-max normalization, and generates transparent, value-grounded explanations. All algorithmic features are integrated into an interactive executive dashboard.

2. MULTI-OBJECTIVE MATHEMATICAL FORMULATION

Standard single-objective optimizations fail in logistics because minimizing cost leads to excessive transit delays, while minimizing time incurs cost premiums. RouteWise AI formulates route selection as a constrained multi-attribute utility problem evaluated over a feasible route space.

2.1 Composite Risk Formulation

Risk is synthesized from three empirical risk dimensions into a unified score R in [0, 1]:

R = 0.50 * P(delay) + 0.30 * R(congestion) + 0.20 * R(geopolitical)

This weighting reflects operational priority: historical delay probability directly degrades SLA compliance, congestion introduces transit variance, and geopolitical risk captures regional chokepoint vulnerability.

2.2 Min-Max Normalization with Singularity Protection

Because shipping cost ($), transit days, and composite risk span disparate measurement scales and dimensions, each metric M in {Cost, Time, Risk} is mapped into a dimensionless unit interval [0, 1] relative to the alternatives available for that specific shipment:

M_norm = (M - M_min) / (M_max - M_min) if (M_max > M_min) else 0.0

Crucially, if all alternatives share identical values (M_max == M_min), the normalized value collapses to 0.0, completely preventing zero-division errors without distorting objective contributions.

2.3 Weighted Composite Scoring Formulation

The overall optimization score S(r) for feasible route r is computed as:

Score(r) = w_cost * Cost_norm(r) + w_time * Time_norm(r) + w_risk * Risk_norm(r)

Subject to: w_cost >= 0, w_time >= 0, w_risk >= 0, and w_cost + w_time + w_risk = 1.0

Lower scores indicate superior routes. Default enterprise baseline weights are w_cost = 0.40, w_time = 0.35, w_risk = 0.25.

3. HARD CONSTRAINT ENFORCEMENT & INFEASIBILITY HANDLING

A critical invariant of RouteWise AI is that hard business constraints are non-negotiable filters. Optimization scoring is strictly restricted to routes satisfying all three hard constraints simultaneously:

- **Budget Constraint:** Shipping Cost <= Max Allowed Shipment Budget ($)
- **Transit Constraint:** Transit Time (days) <= Max Permissible Transit Days
- **Risk Constraint:** Delay Probability <= Max Acceptable Delay Threshold

3.1 Infeasible Route Diagnostic & Closest Alternative Recovery

When market constraints are overly tight and no feasible route exists, RouteWise AI rejects arbitrary fallback choices. Instead, it triggers an intelligent diagnostic that quantifies the normalized constraint violation distance for each candidate:

D_viol = max(0, (Cost - Budget)/Budget) + max(0, (Time - Limit)/Limit) + max(0, (Delay - MaxDelay)/MaxDelay)

The system isolates the route with min(D_viol), generates plain-language violation breakdowns, and identifies the exact budget, time, or risk concessions required to restore feasibility.

4. EMPIRICAL BASELINE EVALUATION & BENCHMARK COMPARISON

To rigorously validate the multi-objective approach, RouteWise AI was benchmarked against three industry baseline heuristics across the complete dataset of 40 synthetic global shipments (153 total route alternatives):

Strategy	Feasible (%)	Avg Cost ($)	Avg Transit	Avg Delay	Avg Opt Score
Weighted Optimization	**85.0%**	**$9,312.09**	**23.9 d**	**16.5%**	**0.1808**
Cheapest Route	85.0%	$8,380.06	26.4 d	19.1%	0.2283
Fastest Route	85.0%	$9,796.56	22.9 d	16.6%	0.2186
Lowest-Risk Route	85.0%	$9,696.31	24.3 d	14.8%	0.2374

4.1 Key Benchmark Insights

1. Superior Composite Balance: Weighted Optimization achieves an Average Optimization Score of 0.1808, outperforming Cheapest (0.2283), Fastest (0.2186), and Lowest-Risk (0.2374) across the network.
2. The Hidden Cost of Single-Minded Baselines: The Cheapest heuristic reduces expenditure to $8,380.06 but incurs a 10.5% transit penalty (26.4 days) and elevated delay probability (19.1%). Conversely, Fastest route achieves 22.9 days but inflates shipping expenditure to $9,796.56.
3. Pareto Efficiency: RouteWise AI balances cost ($9,312.09) and speed (23.9 days) while holding delay to 16.5%, delivering optimal multi-attribute utility without violating enterprise constraints.

5. SENSITIVITY ANALYSIS & ROBUSTNESS PROFILING

Sensitivity analysis examines recommendation stability under weight shifts and constraint variations. A 66-point weight simplex grid (step=0.10, w_c + w_t + w_r = 1.0) was executed on representative shipment corridors:

- **Decision Stability:** For corridor Singapore -> Santos, route SH-001-R01 demonstrated strong robustness, capturing 77.3% of all explored weight configurations.
- **Pareto Switching Boundaries:** Route recommendations remain stable within distinct weight regions; switching occurs sharply at critical frontiers (e.g., when cost weight exceeds 0.65 or transit weight surpasses 0.55), confirming predictable model governance.
- **Constraint Elasticity:** Varying budget from 0.5x to 2.0x demonstrates that loosening budget by 20% increases candidate route availability by 35%, whereas tightening transit time by 15% eliminates 60% of viable candidates.

6. SYSTEM ARCHITECTURE & ENGINEERING STANDARDS

RouteWise AI is structured with clean separation of concerns across 7 core modules:
- Data Layer (src/data_generator.py, data_loader.py): Seeded stochastic generator, schema validation, zero data leakage.
- Preprocessing (src/preprocessing.py): Type casting, composite risk formulation, intermediate hub parsing.
- Constraint Engine (src/constraints.py): Hard constraint filtering, margin calculation, closest-infeasible recovery.
- Optimization Engine (src/optimizer.py): Min-max scaling, weighted sum scoring, division-by-zero protection.
- Explainability & Recommender (src/recommender.py, explainability.py): Transparent breakdowns, trade-off comparisons.
- Sensitivity & Evaluation (src/sensitivity.py, evaluation.py): Simplex grid search, baseline benchmark comparisons.
- Presentation Layer (app.py): Executive Streamlit dashboard, Plotly radar/box visualizations, interactive scenario modeling.

7. LIMITATIONS, HEURISTICS & PRODUCTION ROADMAP

Model Assumptions & Boundaries:
1. Heuristic Nature: Weighted-sum scoring is a decision-support heuristic, not a mathematical guarantee of global logistics optimality.
2. Relative Normalization: Scores are normalized relative to available alternatives per shipment and should not be compared across unrelated shipments.
3. Linear Trade-off Assumption: Real-world supply chain penalties are non-linear (e.g., SLA breach step-penalties).

Production Roadmap:
- Phase 1: Real-time telemetry ingestion (AIS maritime vessel tracking, weather APIs, port congestion indices).
- Phase 2: Dynamic Non-linear Optimization (Pareto Front exploration via NSGA-II / Epsilon-constraint solvers).
- Phase 3: Automated carrier contract rate bidding and dynamic SLA penalty mitigation.